CLASS PROJECT U1 – U2

Three targets, 194 realities.

Executive EDA Report

TecNM Campus Ciudad Guzmán

Master of Science in Computer Science

Course: Selected Topics

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WHO TRIPLE BILLION TARGETS · 2018–2025

Three Goals, 194 Realities

Only one of three targets was met

HEALTHIER POPULATIONS

1,750 M

175% of target · Target met

Marker = 1,000 M target

Driven by clean household fuels, sanitation and air quality.

HEALTH EMERGENCIES PROTECTION

698 M

70% of target · Off track

Marker = 1,000 M target

The most irregular target, with setbacks in 2021 and 2023.

UNIVERSAL HEALTH COVERAGE

567 M

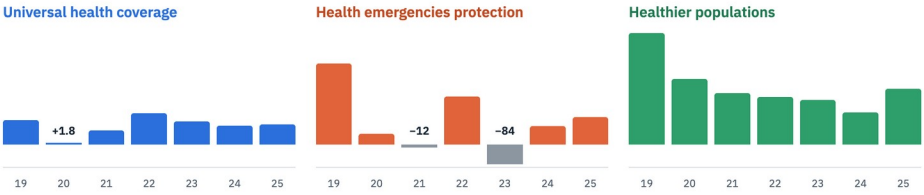
57% of target · Off track

Marker = 1,000 M target

Progress led mainly by HIV treatment coverage.

The pace was not constant

People gained each year, global (M) · grey = setback



Almost stalled in 2020 (+1.8 M vs. +105 M in 2019), during COVID-19.

Fell back twice: −12 M in 2021 and −84 M in 2023.

Never fell back; gains every year since 2018.

Progress varies by region

Contribution in 2025 (M people vs. 2018)

Region	UHC	HEP	HPOP
South-East Asia	255	445	**839**
Western Pacific	124	96	536
Africa	84	158	124
Eastern Mediterranean	40	30	93
Europe	37	36	85
Americas	27	−68	72

The Americas are the only region that lost ground, in emergency protection. Large regions add more in absolute terms.

What helps, what hurts

Top global indicators in 2025 (M people vs. 2018)

+626	Clean household fuels
+468	Safely managed sanitation
+344	IHR emergency capacities
−184	Adult obesity
−50	Child obesity
−20	Financial protection (catastrophic spending)

Environmental gains lead; obesity is the largest setback worldwide.

Spotlight: Mexico

Steady gains in healthier populations, a pandemic dip in coverage, and the largest emergency-protection setback of all 194 countries.

M people vs. 2018

HEALTHIER POPULATIONS

+18.6

Grew every year, from 4.4 M in 2019. Led by safely managed sanitation (+14.7 M).

UNIVERSAL HEALTH COVERAGE

+5.3

Below 2018 levels in 2019–2021, lowest in 2020 (−4.3 M). Recovered from 2022.

EMERGENCIES PROTECTION

−24.1

Negative every year. Lowest of 194 countries, ahead of the US (−21.7 M) and Brazil (−12.5 M).

Lines: 2018–2025 on a shared scale · dashed line = 2018 level

ADDING THE MOST

+14.7	Safely managed sanitation
+5.2	Detect and respond to emergencies

SUBTRACTING THE MOST

−18.4	Prevention (high-threat pathogen vaccination)
−11.0	IHR emergency capacities (e-SPAR)

THREE TARGETS, 194 REALITIES.

1. Executive summary

1.1 Problem context

The Triple Billion project is a strategic initiative of the World Health Organization and the core of its Thirteenth General Programme of Work (GPW13). The programme was defined for 2019–2023, with 2018 as the baseline year. However, in 2022 the WHO extended it to 2025 due to the disruptions caused by COVID-19.

The targets expected by 2025 are one billion more people in each of three areas:

- **Universal health coverage (UHC):** one billion more people with access to health services without financial hardship.
- **Health emergencies protection (HEP):** one billion more people better protected against emergencies such as pandemics and outbreaks.
- **Healthier populations (HPOP):** one billion more people enjoying better health and well-being.

Measuring progress shows whether public health efforts are working and in which areas there are gaps. The year 2025 is the point at which the targets are evaluated, marking the close of GPW13 and the start of GPW14 (2025–2028), which aims to accelerate progress towards the 2030 Sustainable Development Goals. According to the WHO, the world is far from achieving the health-related SDGs by that year [1].

1.2 Data analyzed

Source: World Health Organization (WHO), Triple Billion dashboard of the WHO Data portal. The current statistical release is dated May 18, 2026 and is distributed under a CC BY 4.0 license [1].

Download date: 09/30/2026

Overview: The file has 62,712 records and 8 variables, with annual data from 2018 to 2025. It covers 194 countries, the 6 WHO regions and the global total. It includes 39 contribution indicators: 17 for UHC, 4 for HEP and 18 for HPOP, each group with one indicator that adds up the billion total.

Nature of the data: The data are WHO estimates calculated from information reported by countries and supplemented with statistical models. This is why they reflect events such as the COVID-19 pandemic and include 90% uncertainty intervals. For each indicator, the WHO fits a model to the 2000–2019 trend and adjusts it with data observed during the pandemic, for the indicators that have such data [2].

1.3 Main variables

Table 1. Main variables of the dataset.

Variable	Data type	Description
Area	Text (object)	Geographic level: country, WHO region or global total
Name	Text (object)	Name of the country, region or global total
Year	Integer (int64)	Year of the estimate, from 2018 to 2025
Billion	Text (object)	Target it contributes to: uhc, hep or hpop
Tracer	Text (object)	Indicator that generates the contribution
Mean_value	Decimal (float64)	Central estimate of the change in number of people relative to 2018
Lower_value	Decimal (float64)	Lower bound of the 90% uncertainty interval
Upper_value	Decimal (float64)	Upper bound of the 90% uncertainty interval

A positive value indicates people gained relative to 2018, for example, more people with safely managed sanitation. A negative value indicates a setback, for example, more adults with obesity. A zero corresponds to the baseline year or to indicators that do not apply, have no data or have no assessable trend.

The dataset is in long format: each row is a unique combination of location, year and indicator. It has no missing values or duplicates, and each of the 201 locations has 312 records (8 years × 39 indicators).

1.4 Relevance for decision-making

The analysis is useful for governments, ministries of health and international organizations that must evaluate the close of GPW13 and plan GPW14.

It makes it possible to identify which targets are lagging, which indicators detract from progress and which regions or countries are moving backwards. This helps decide where to invest and which policies to prioritize. The WHO itself notes that progress in UHC depends mainly on HIV coverage and that childhood immunization and tuberculosis are still recovering from the pandemic.

It is relevant for Mexico because it has the lowest value of the 194 countries in emergency protection and because obesity detracts from progress in population health.

THREE TARGETS, 194 REALITIES.

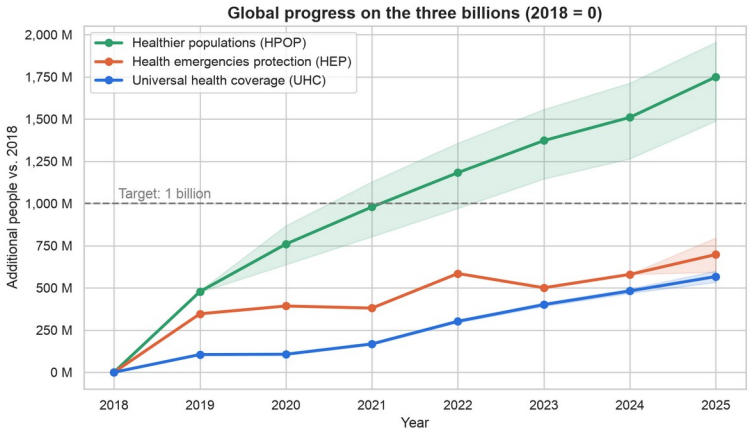
2. Exploratory Data Analysis

This section explores progress towards the Triple Billion targets between 2018 and 2025 from four perspectives: the global total, the WHO regions, the countries and the indicators, with a case study on Mexico. It analyzes trend, seasonality, cyclical behavior, outliers and pattern changes over time. Unless otherwise stated, figures correspond to the central estimate (Mean_value) and are expressed in millions of people relative to 2018.

2.1 Global overview against the target

To assess progress towards the targets, the cumulative global contribution of each billion between 2018 and 2025 was plotted (Figure 1). Each line represents the central estimate, the shaded band the 90% uncertainty interval and the dotted line the target of 1,000 million people.

Figure 1. Global progress of the three billions against the target, 2018–2025.

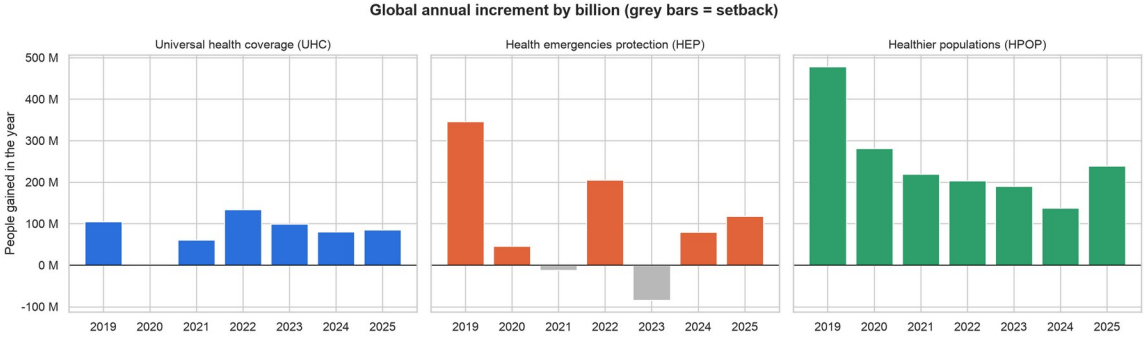


All three billions show an upward trend, although at very different paces. HPOP crosses the target between 2021 and 2022 (from 979 to 1,183 million) and ends 2025 at 1,750 million. HEP and UHC remain below the target throughout the period.

2.2 Changes over time

Since the values are cumulative, the annual increase of each billion was calculated to analyze the pace of progress (Figure 2). Grey bars indicate setbacks.

Figure 2. Annual global increase by billion, 2019–2025.



In UHC, the increase fell from 105 million in 2019 to just 1.8 million in 2020; from 2021 onwards it recovered, peaking in 2022 at 134 million.

HEP is the most irregular billion. After adding 346 million in 2019, it fell back in 2021 (−12 million) and in 2023 (−84 million), with years of strong progress in between, such as 2022 (205 million).

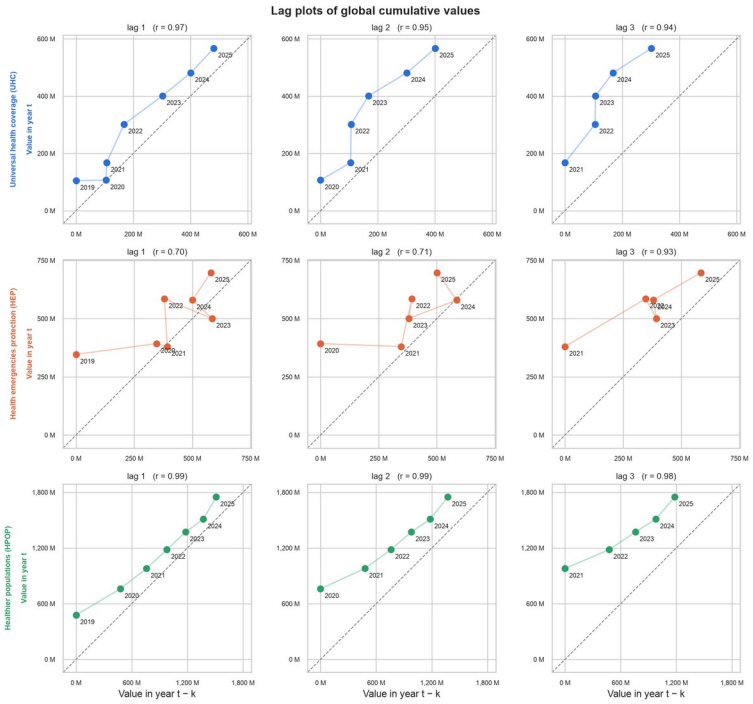
HPOP never falls back: its increase gradually declines from 478 million in 2019 to 138 million in 2024 and rebounds to 239 million in 2025.

The slowdown in UHC in 2020 coincides with the COVID-19 pandemic, which disrupted essential services such as immunization and tuberculosis treatment. The setbacks in HEP show that emergency protection does not progress in a sustained way.

2.3 Autocorrelation: lag plots

To assess the persistence of the series and the possible presence of seasonality or cycles, lag plots were built (Figure 3). In each panel, the X axis shows the value for year $t - k$ and the Y axis the value for year t. The dotted diagonal represents $y = x$, so a point above it indicates growth and one below it, a decrease.

Figure 3. Lag plots (k = 1, 2 and 3) of the global cumulative values by billion.



In HPOP and UHC the points form an almost straight line above the diagonal, with autocorrelations between 0.94 and 0.99, which confirms a persistent upward trend. The exception is UHC in 2020, which at lag 1 sits practically on the diagonal.

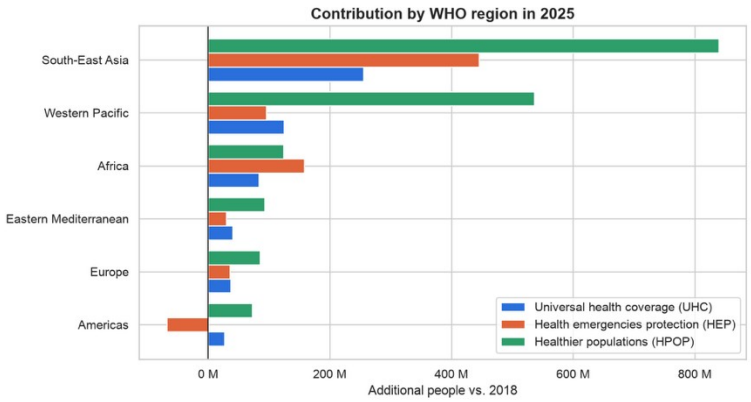
HEP is the most scattered series. The points for 2021 and 2023 fall below the diagonal because the global value decreased in those years: from 393 to 380 million and from 585 to 501 million.

Seasonality cannot be observed, since the data are annual; this pattern only appears with monthly or quarterly data. Nor are there circular shapes or separate clusters that would suggest cyclical behavior, although eight years are not enough to rule it out.

2.4 Comparison by WHO region

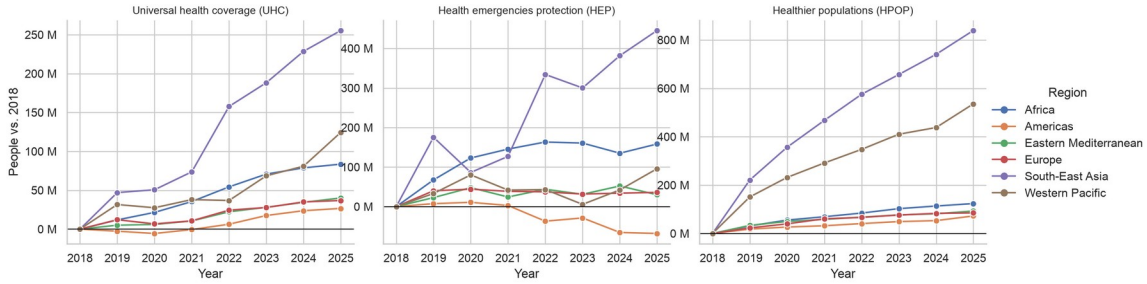
The contribution of the six WHO regions in 2025 (Figure 4) and their evolution between 2018 and 2025 (Figure 5) were compared.

Figure 4. Contribution by WHO region and billion in 2025.



South-East Asia is the region that contributes the most across all three billions: 839 million in HPOP, 445 in HEP and 255 in UHC. It is followed by the Western Pacific in HPOP (536 million) and UHC (124 million), while Africa stands out in HEP, with 158 million. The Americas show the lowest progress in UHC (27 million) and HPOP (72 million), and are the only region with a negative value: −68 million in HEP.

Figure 5. Evolution of the contribution by WHO region, 2018–2025.



In HPOP all regions grow steadily. In UHC, the Americas remain below zero in 2019 and 2020 before recovering. HEP is the most irregular panel: South-East Asia drops from 175 to 86 million between 2019 and 2020 before soaring, and the Americas turn negative from 2022 onwards.

The differences largely reflect the population size of each region, since the values are absolute: a large region contributes more without that implying better relative performance. Even so, the sustained setback of the Americas in HEP is a relevant finding for the region.

2.5 Countries and outliers

To analyze the distribution across countries, a box plot was built with the 194 countries in 2025 (Figure 6). A symmetric logarithmic scale, which accepts negative values, was used because contributions range from −24 to +724 million. Outliers were identified with the interquartile range (IQR) rule: values below Q1 − 1.5·IQR or above Q3 + 1.5·IQR.

Figure 6. Distribution of the contribution by country and billion in 2025.

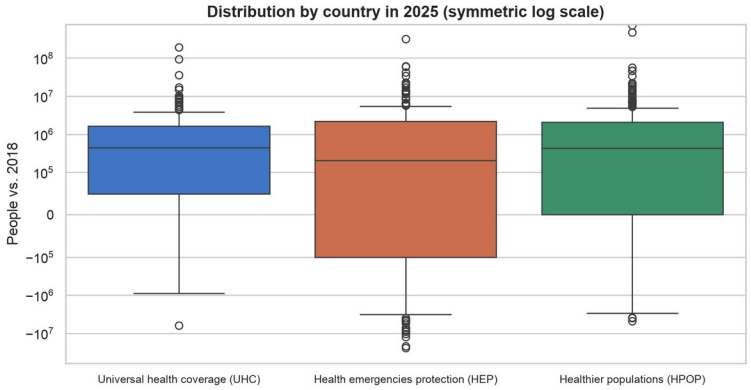


Table 2. Outliers and countries with the highest and lowest contribution in 2025

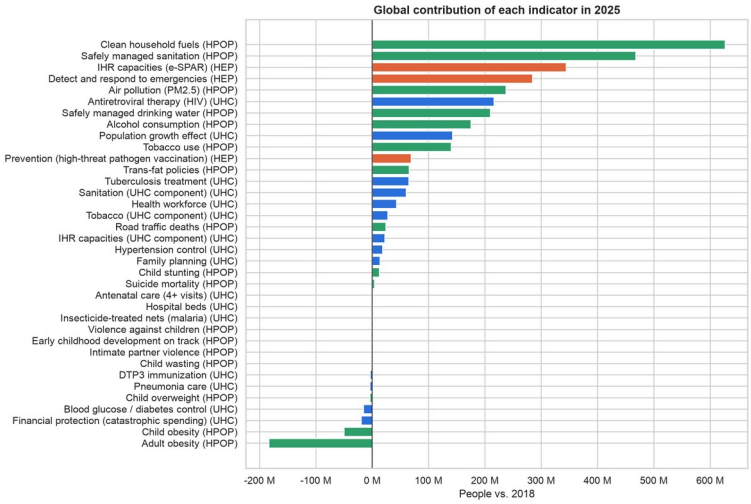
Billion	No. of outliers	Highest contribution	Lowest contribution
UHC	20	India (194 M)	Brazil (−6.2 M)
HEP	36	India (318 M)	Mexico (−24.1 M)
HPOP	34	India (724 M)	Iran (−4.8 M)

The distribution is highly skewed. Medians are small (between 0.2 and 0.4 million), while means are much larger (from 2.9 to 9.0 million), because a few very populous countries pull them up. HEP is the only billion whose box clearly drops below zero: at least a quarter of the countries (Q1 = −102 thousand people) fell back in emergency protection.

2.6 Contribution by indicator

To identify which indicators drive or hold back progress, the global contribution of each component in 2025 was plotted, excluding the totals of each billion (Figure 7).

Figure 7. Global contribution of each indicator in 2025.

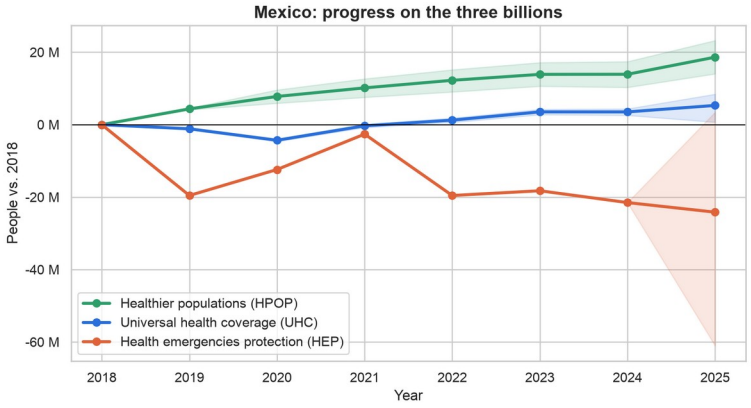


The indicators that add the most are clean household fuels (626 million), safely managed sanitation (468 million), International Health Regulations capacities (344 million), emergency detection and response (284 million) and air quality (237 million). At the opposite end, adult obesity subtracts 184 million, followed by child obesity (−50 million), financial protection against catastrophic spending (−20 million) and glucose control (−16 million).

2.7 Mexico case

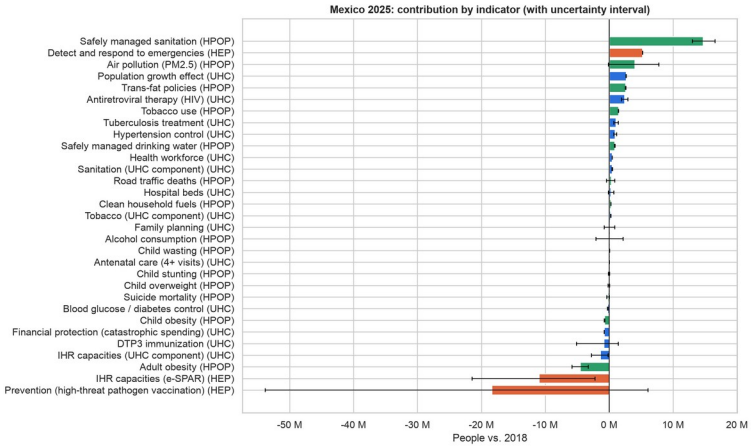
The case of Mexico was analyzed because of its national relevance. Figure 8 shows its evolution by billion and Figure 9 the contribution of each indicator in 2025, with error bars representing the 90% uncertainty interval.

Figure 8. Evolution of Mexico across the three billions, 2018–2025.



HPOP grows steadily, from 4.4 million in 2019 to 18.6 million in 2025. UHC remains below zero between 2019 and 2021, with a low of −4.3 million in 2020, and recovers to 5.3 million in 2025. HEP is negative in every year and ends 2025 at −24.1 million, the lowest value of the 194 countries.

Figure 9. Contribution of Mexico by indicator in 2025, with 90% uncertainty interval.



The indicators that add the most are safely managed sanitation (14.7 million), emergency detection and response (5.2 million) and air quality (4.0 million). Those that subtract the most are prevention in HEP (−18.4 million), e-SPAR capacities (−11.0 million) and adult obesity (−4.5 million).

Adult obesity and e-SPAR capacities are clear setbacks, since their entire interval is negative (from −5.8 to −3.3 million and from −21.5 to −2.3 million, respectively). In contrast, the interval for prevention ranges from −54 to +6 million and crosses zero, so the HEP total (from −61.0 to +3.4 million) is inconclusive: the data suggest a setback, but do not allow it to be stated with certainty.

THREE TARGETS, 194 REALITIES.

3. Conclusions

The exploratory analysis shows that progress towards the Triple Billion targets between 2018 and 2025 was real, but uneven. Globally, only the healthier populations (HPOP) target was met, with 1,750 million people, driven by environmental improvements such as the use of clean household fuels, safely managed sanitation and air quality. In contrast, health emergencies protection (HEP) reached 698 million and universal health coverage (UHC) 567 million, which is consistent with the WHO assessment that the world is not on track to meet these two targets (Figure 1).

Behavior over time also differs between targets. All three series show an upward trend and show no seasonality or cycles, which is to be expected with annual data covering only eight years (Figure 3). However, the pace was not constant: progress in UHC practically stalled in 2020, coinciding with the COVID-19 pandemic, and HEP fell back in 2021 and 2023 (Figure 2). This suggests that emergency protection is the target most vulnerable to crises and the one that progresses least steadily.

The sharpest differences appear when comparing regions and countries. South-East Asia and the Western Pacific account for most of the progress, while the Americas are the only region with a setback, in HEP (Figures 4 and 5). Across countries, the distribution is highly skewed: a few very populous countries, such as India and China, explain much of the global progress, while at least a quarter of the countries fell back in emergency protection (Figure 6). For this reason, global figures can hide what happens in most countries. At the indicator level, obesity is the main setback, both worldwide and in Mexico (Figures 7 and 9).

Mexico reflects this inequality. It progresses steadily in population health (18.6 million) and, after falling back during the pandemic, recovers in universal coverage (5.3 million). However, it records the lowest value of the 194 countries in emergency protection (−24.1 million), although its uncertainty interval does not allow that setback to be confirmed (Figures 8 and 9).

In short, three common targets produced 194 different realities. The differences observed could be related to political, economic, social and cultural factors in each country, which are not part of this dataset. Incorporating them in future analyses, together with population-relative indicators, would help explain why some countries progress and others fall back.

THREE TARGETS, 194 REALITIES.

4. References

[1] World Health Organization. (n.d.). *WHO Triple Billion progress data (contributions)*. Data.who.int. https://data.who.int/dashboards/global-progress/triple-billion?m49=001

[2] World Health Organization. (n.d.). *Triple Billion forecasting methods*. https://www.who.int/publications/m/item/3b-forecasting-methods

THREE TARGETS, 194 REALITIES.

5. Appendices

https://github.com/scarlette19991989/Three-targets-194-realities